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SEAT No. :

P5123

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B.E. / Insem. - 530

B.E. (Mechanical Engineering)

OPERATIONS RESEARCH

(Elective - II) (2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of electronic pocket calculator is allowed.
- 4) Assume suitable data if necessary.

Q1) Maximize:

$$Z = 3x_1 + 5x_2 + 4x_3$$

[10]

Subject to constraints - $2x_1 + 3x_2 \leq 8$

$$2x_2 + 5x_3 \leq 10$$

$$3x_1 + 2x_2 + 4x_3 \leq 15$$

$$x_1, x_2, x_3 \geq 0$$

OR

Q2) a) Write short notes on :

[4]

i) Laplace Criterion

ii) Hurwicz Criterion

b) Daily demand for litres of milk at a supermarket is given by following probability distribution :

[6]

Daily demand	100	150	200	250	300	350	400
Probability	0.20	0.25	0.30	0.10	0.05	0.05	0.05

If milk is not sold the same day it becomes useless. The supermarket is selling Rs.31 per litre whereas cost per litre to supermarket is Rs.25. If each day's demand is independent of previous day's demand, how many litres of milk should be ordered everyday?

P.T.O

- Q3) a)** Job shop needs to assign 4 jobs to 4 workers. The cost of performing a job is a function of the skills of the workers. Table summarizes the cost of the assignments. Worker 1 cannot do job 3, and worker 3 cannot do job 4. Determine the optimal assignment using the Hungarian method. [6]

Worker	Jobs			
	J1	J2	J3	J4
W1	Rs.50	Rs.50	---	Rs.20
W2	Rs.70	Rs.40	Rs.20	Rs.30
W3	Rs.90	Rs.30	Rs.50	----
W4	Rs.70	Rs.20	Rs.60	Rs.70

- b) Explain Least Cost method of allocation for transportation problem. [4]

OR

- Q4) a)** Find the optimal solution for given transportation problem. [10]

Factories	P	Q	R	S	Availability
A	10	8	7	12	500
B	12	13	6	10	500
C	8	10	12	14	900
Demand	700	550	450	300	

- Q5) a)** Solve the following game using dominance property. [6]

		Player B		
		I	II	III
Player A	I	1	7	2
	II	6	2	7
	III	6	1	6

- b) Explain break even analysis & margin of safety with neat sketch. [4]

OR

Q6) a) A ball bearing manufacturing company is planning to install an additional plant which will require leasing new equipment for monthly payment of Rs.60,000. Variable cost would be Rs.20 per item and each item would retail for Rs.70. **[6]**

- i) How many ball bearing units must be sold in order to break even?
- ii) What would be profit or loss if 1,000 items are made & sold in a month?
- iii) How many items must be sold to realize a profit of Rs.40,000?

b) Write short notes on : **[4]**

- i) Pure and mixed strategies
- ii) Dominance rule

